

ANNUAL REPORT



1994-95



National Academy of Agricultural Research Management
Rajendranagar, Hyderabad - 500 030, India



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Published by:
Dr Jagdish C. Kalla, Director
NAARM, Hyderabad - 500 030.

Printed at:
NAARM Offset Press

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INTRODUCTION

Over the last four decades, Indian Agriculture has recorded remarkable increase in agricultural production, and has acquired considerable resilience and stability, largely through generation of better technology in land and water management, input use, control of pests and pathogens, and adoption of scientific production techniques. This success has underlined the need for a better management of our education, research and transfer of technology systems to optimize production in agricultural sector.

Realizing the need for an institutional set-up to impart training in agricultural research project management, and for adopting principles of educational technology for better education, the National Commission on Agriculture (NCA) recommended the establishment of an all India institute for administration and management in agriculture. The National Academy of Agricultural Research Management (NAARM), earlier known as Central Staff College of Agriculture, was thus set up in 1976 as one of the constituent institutions of the Indian Council of Agricultural Research (ICAR) under the aegis of the Department of Agricultural Research and Education (DARE), Ministry of Agriculture, Govt. of India.

The Academy, since its inception, has been assisting the deemed-to-be universities, research institutes of the ICAR, and the State Agricultural Universities (SAUs) by providing training input for management of agricultural research, education, training and extension.

The Academy is guided by a Management Committee in planning, organizing and conducting various types of training programmes, research studies, and allied activities. The committee reviews and promotes the programmes of the Academy, and approves new areas of activities and suggestions including physical targets. The members of the committee are from the ICAR Headquarters, its institutes, the state governments and eminent scientists and administrators.

The Academy is guided by an Academic Council for planning various academic programmes and activities, and Research Council monitors the research activities of the Academy.

Keeping in view of the responsibilities, the Academy is organized into six units of management specialization. These are: Agricultural Research Systems and Policies, Educational Systems and Technology, Transfer of Technology System and Policies, Project Management, Human Resources Development, and Information and Documentation Units. All the activities of NAARM are supported by centralized services such as Administration and accounts, Farm, Production and publication, Health and hostel services. Presently the Academy has 178 members including all categories of staff and senior and principal scientists.

The Academy is located in the sylvan environs of Rajendranagar, near Andhra Pradesh Agricultural University (APAU), on the south-western periphery of Hyderabad city. The well-planned campus, spreading over 50 hectares, has several modern buildings including academic block, faculty-cum-laboratory-cum-library building, administrative block, halls of residence, scientists' home, agro-technology demonstration blocks, and several faculty and staff residences.

Dr Jagdeesh C. Kalla, Joint Director assumed charge as Director of the Academy on Feb. 25, 1995 in place of Dr C. Prasad.

Major Mandates of the National Academy of Agricultural Research Management

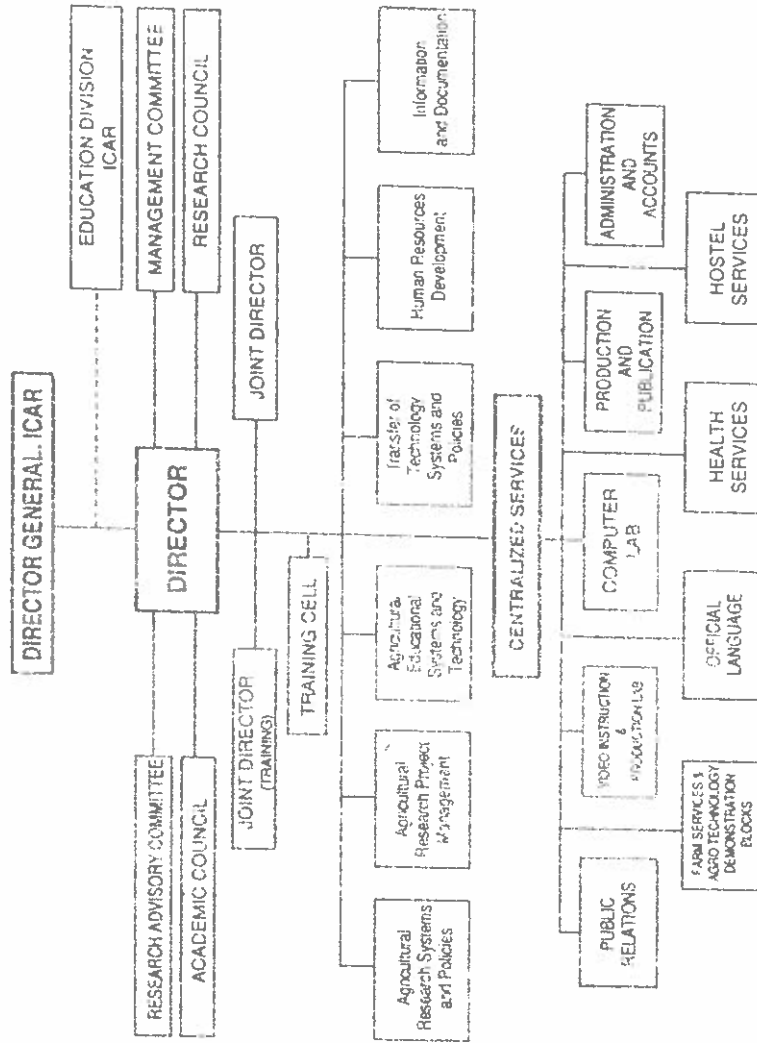
1. To organise foundation courses for new scientist cadre entrants to the Agricultural Research Services of ICAR.
 2. To organise Seminars, Conferences and Workshops on all aspects relating to Agricultural Research Management.
 3. To organise and develop training programmes in Agricultural Research Management for senior and middle level scientists of ICAR Institutes.
 4. To build up high quality resource material in the subject of Agricultural Research Management based on actual field experience.
 5. To undertake a systematic review and study of management problems of ICAR Research Institutes, Agricultural Universities, Agricultural Research Projects and Systems.
 6. To organise, liaise and coordinate programmes of international co-operation in the fields of Agricultural Research Management.
 7. To undertake a programme of publishing high quality training material related to Agricultural Research Management.
 8. To plan, organise and conduct training programmes, workshops and seminars in Educational Technology, Curriculum development and structural learning for the faculty of Agricultural Universities and ICAR Institutes, and
 9. To function as a repository of ideas and information, both national and international in the field of Agricultural Research Management and Education, and develop as a clearing house for dissemination.
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Academy is organized
in: Agricultural Re-
search Technology, Trans-
mission, Human Re-
lation Units. All the
units such as Admin-
istration, Health and hos-
pital including all cat-

of Rajendranagar,
in the south-western
region spreading over 50
hectare block, faculty-
rooms, halls of residence,
libraries, and several fac-

in charge as Director
General.

ORGANIZATIONAL STRUCTURE OF NAARM



HIGHLIGHTS

An achievement is a sum total of many attributes emanating out of a concerted team effort in the directions of fulfilment of mandate. Each step forward leads to greater quest for acquisition of knowledge, skill and attitudes of individuals aided by enterprising environment and effective leadership that ensures success and achievement. Highlights of this year are listed here.

1. One of the hallmarks during the period under report was successful conduct of an International training workshop on "Management of Agricultural Research" for the Directors of Agricultural Research Institutes of Nigeria from November 21 to December 2, 1994.
2. Once again, Raj Bhasha Rolling Shield was awarded to the Academy in recognition of the distinguishable work done in implementing the official language policy of the union.
3. The Academy received an appreciation from the Financial Advisor, ICAR (DARE), New Delhi for the best performance in settling the audit observation, and was cited as an example to be followed by the ICAR Institutions.
4. A three-week summer institute on "Recent Advances in Agricultural Education Technology" was organized from May 7 to 28, 1994. Dr Ahmed, DDG (Education), ICAR, New Delhi inaugurated the programme.
5. The Union Minister of State for Personnel & Public Grievances, Mrs. Margaret Alva visited the Academy, and delivered the valedictory address to the participants of the two-week national training workshop on "Human Resources Management for Women Scientists" on Sept.8, 1994.
6. Hon'ble Governor of Andhra Pradesh, Krishan Kant visited the Academy, and inaugurated the three-day national workshop on "Non-Pesticidal Approach to Pest Management - A New Direction" on Sept.20, 1994. The workshop was organized in collaboration with the Dept. of Agriculture, Govt. of Andhra Pradesh, Centre for World Solidarity (CWS), and OXFORD (India). Minister of Agriculture, Govt. of A.P., Mr K. Janareddy attended the "Kisan Gosthi" and delivered the valedictory address to the participants.

HOSTEL
SERVICES

HEALTH
SERVICES

OFFICIAL
LANGUAGE

FARM SERVICES &
AGRO-TECHNOLOGY
DEMONSTRATION
BLOCKS

7. A mobile seminar on "Management of Training - Local Institutional Resources and Experiences" was organized at the Academy on June 27, 1994 to create awareness on various physical facilities, understand the innovations and pedagogy of training, know the experience available, and to chalk-out a strategy for linkages between various agricultural institutions in and around Hyderabad.
 8. A three-day regional conference of "Peasant Women of Southern States in Agriculture, Environment and Rural Development" was organized at the Academy on Oct. 27, 1994 in collaboration with the Centre for Women's Development Studies (CWDS) and International Federation for Women in Agriculture (IFWA) with an objective to promote greater interaction between peasant women and scientists, technologists, and representatives of Govt. agencies involved in framing policies for agriculture, environment and general rural development.
 9. Faculty Development Programme in Educational Technology was organized at C.C.S. Haryana Agricultural University, Hissar from Jan. 12 to 25, 1995. The programme was initiated at the instance of Director, Academy of Agricultural Research and Education Management, CCS Haryana Agricultural University, Hissar. A total of 27 participants were trained in the programme. Similar programme was also organised at Dr Y.S. Parmar University of Horticulture and Forestry, Solan, HP from July 2 to 8, 1994. A total of 24 participants were trained in various aspects of educational technology and research management in this programme.
 10. A four-day national workshop on "Manpower Development in Extension Education for Meeting Future Challenges of Agriculture" was organized in collaboration with the Indian Society of Extension Education (ISEE) from Feb. 13 to 16, 1995. Dr M. Channa Reddy, His Excellency the Governor of Tamilnadu inaugurated the workshop.
 11. The Agricultural Universities Information System (AGRIUNIS) Project was concluded by March, 1995, and certain comprehensive recommendations were made. The study is, on the whole, a valuable and timely contribution to agricultural education and research institutions in the country.
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ACTIVITIES

The Academy organized various activities related to training, research, education, workshops and consultancy to fulfill the needs of the clientele groups working at various levels of management in the agricultural research and development organizations, and each of these activities is described below:

TRAINING

Training is an important input for Human Resources Development in all categories of human resource-mix engaged in teaching, research and extension functions. Agricultural research and education management is one such important sub-sectors. The Academy imparts training in the areas of research management, educational technology, and training management for various levels of scientists of the ICAR and the SAUs, which form a part of the National Agricultural Research System (NARS). It also conducts training programmes for executives, senior research managers, and administrative and accounts officers. The training programmes have been grouped into two main sub-heads viz., Foundation course for Agricultural Research Service (FOCARS) for probationers, Specialized and advanced training programmes including workshops, seminars, and conferences for senior level scientists and policy makers.

Foundation Course

Foundation course is designed to the scientists recruited to Agricultural Research Service of the ICAR. It is the most important programme of the Academy. The course aims at providing exposure to the trainees on the concepts and principles of project management with special emphasis on project formulation and implementation. It also includes instruction in related areas on human resources development and management, transfer of technology systems and policies, and information management. The specific objectives of the FOCARS are as follows:

Objectives

1. To acquaint the scientist - trainees with the general agricultural scenario in the country with specific reference to research, education, extension education and training.

2. To make them aware of the global agricultural scenario in general and International agencies devoted to agricultural development in particular.
3. To orient the scientist - trainees in the administrative and management aspects of the Indian Council of Agricultural Research including Agricultural Research Service, Agricultural Scientists Recruitment Board, functioning of various ICAR bodies, personnel policy, finances and monitoring.
4. To expose them in agricultural management in relation to research identification, prioritisation, project formulation, implementation and management.
5. To equip the scientist - trainees with the techniques of research identification, prioritisation, project formulation, implementation and management.
6. To give an insight into personality development, leadership styles, creativity, problem solving techniques, group processes and conflict management.
7. To provide an understanding and skills in information and communication technology management including computer application, communication aids and techniques, Video programme production, skills for oral communication, technical report writing and presentation.
8. To give the trainees an understanding of and an insight into the socio-economic structure of rural society as well as farming community.
9. To expose scientist - trainees to the outside world through special scientific and general lectures and seminars including social, cultural, moral and spiritual values and commitment to work as well as creation of a healthy work environment.

The course is structured into three phases, each covering a duration of seven weeks, and extending to a total period of six months. The first phase of the training is at the Academy, and concentrates mainly on the basics of Agricultural Research Management, Organizational Behaviour, Transfer of Technology, Information Management, Administration and Ac-

counts. Procedures. The second phase, Field Experience Training (FET) is held at regional stations of National Agricultural Research Project (NARP) of the ICAR Institutes and State Agricultural Universities (SAUs). The FET is designed to impart training in project identification and formulation. This phase helps the trainees to develop an appreciation for rural life and environment. In the third phase of training at the Academy, the trainees are given an exposure in those areas which would strengthen their skills in scientific writing, and presentation and improvement of communication skills.

The overall programme also includes educational visit to various research institutes in and around Hyderabad to familiarize them with different organizations and the research activities. During the period under report, 70 ARS Scientists, freshly recruited into the service, completed their training. Currently 93 ARS Scientists are receiving training. The details are as follows:

S.No.	Title of the Programme	Dates	Co-ordinators	No. of participants
1.	FOCARS 49	25.7.94 10.1.95	Dr VE.Sabarathnam Dr K. Venkateswarlu	70
2.	50	13.2.95 13.7.95	Dr S. Shanmugam Dr T. Balaguru	46
3.	51	20.2.95 20.7.95	Dr S.N. Saha Dr D. Rama Rao	47

Specialized and Advanced Courses

Research management training is a continuous process of learning and updating skills. The Academy offers various specialized programmes to all categories of scientists. These specialized programmes essentially aimed at developing an appreciation and providing adequate skills to various levels of scientists of the ICAR and SAUs. The trainees of senior courses include Principal Scientists, Professors, Heads of Divisions/Departments,

Directors of Research, Project Coordinators, Directors of institutes and Deputy Directors-Generals. During the year under report, twelve senior level programmes were offered which are listed in a table, and a brief account of the programmes is as follows:

1. Agricultural Research Project Management

This course exposed the research managers to applications of management concepts to agricultural research projects for their efficient implementation and to make them relevant. Various aspects of research project cycle, and other related topics like management of differences and brainstorming are covered through information sharing, group discussion, case analysis and exercises.

2. Management Information System for Agricultural Research

The course provided guidelines on the methodology for the development of computer-based Management Information System (MIS) and Project Budgeting System (PBS) that would ultimately help managers to take timely and meaningful decision on various aspects of research projects, human resources and financial resources. The course enabled the participants to develop skills through project work, case studies, and hands-on experience with personal computers.

3. Computer Applications in Agricultural Research

This course was aimed at providing skills to the senior scientists of National Agricultural Research Project (NARP) centres of State Agricultural Universities of Southern, Eastern and Western Zones, on the use of Computers and different software packages for processing of data, and presentation of information. Various user-friendly packages related to word processing, statistical analysis and spreadsheets were covered. A brief account of computer virus and vaccines was also provided. Major emphasis was on hands-on experience with the computers for the use of various packages. This training programme was sponsored by World Bank.

4. Fisheries Extension Education

The orientation course was organized for the State fisheries department extension officers and assistants as a partial fulfilment of the course conducted by the Central Institute of Fisheries Education.

10

S No.	Name of the programme	Dates	Co-ordinator(s)	No. of participants
1	Course on Agricultural Research Project Management	Apr.4-16, 1994	Dr S.N. Saha	17
2	An Advanced Course on Management Information System for Agricultural Research	Apr.19-21 1994	Dr T. Balaguru	13
3	FDP in Educational Technology at YSPUHF	July 2-8 1994	Dr A.Gopalam Dr Dinabandhu	24
4	Course on Computer Applications in Agricultural Research	July 19-29 1994	Mr U. Muralidhar	30
5	Orientation Course on Fisheries Extension Education	Aug. 23-31 1994	Dr A.U. Gadewar Dr S. Shanmugam	14
6	Course on Human Resources Management for Women Scientists	Aug. 29-Sept. 8 1994	Dr M.M. Anwer Dr S. Shanmugam	8
7	Course on Production of Educational Video Materials	Nov.14-26 1994	Mr K.R. Prabhakar Mr. Janardhan Rao	20
8	Course on Management for Improving and Assessing Agric. Research Productivity	Nov.21-Dec.17 1994	Dr V.Raghavendra Rao Dr P. Kumar Dr VE. Sabarathnam	12
9	Course on Training of Trainers	Dec.6-17, 1994	Dr T. Balaguru Dr P. Manikandan	14
10	FDP on Educational Technology at Dr.R.S. Dalal HAU, Hisar	Jan 12-24 1995	Dr A. Gopalam	28
11	Course on Management Information System for Agricultural Research	Feb.7-18 1995	Dr P. Manikandan Dr T. Balaguru Dr J.C. Kalla	20
12	Course on Computer Applications in Agricultural Research	Mar.21-31 1995	Mr U. Muralidhar Dr K. Vidyasagar Rao	22

The course created an awareness about concepts, philosophy and principles of extension, communication, social change, socio-psychological behaviour of the target group, group efforts and programme evaluation methods in Fisheries Extension. It also provided the skills of dealing and managing human resources at operational level, and provided exposure to historical perspectives of fisheries extension, problems and prospects in fisheries sector and participatory information collection tools. The course also provided hands-on experience in audio-visual aids, instructional plans and curriculum development.

5. Human Resources Management for Women Scientists

A two-week national training workshop was organized for women scientists to expose them to the concept of human behaviour in organizations and to examine the issues of management of people in agricultural research, thereby, increase the productivity of research.

The topics covered included organizational behaviour; values and attitudes; personality and stress; perception and motivation; fundamental of group behaviour; group dynamics; leadership; conflict management; organizational structure; performance evaluation and rewards, etc.

The Union Minister of State for personal and public grievances, Mrs. Margaret Alva facilitated the conduct of valedictory function of the workshop. During her speech, she stressed on the applicability of science and technology to the local needs, and called for demystifying science and technology in order to bring attitudinal change among people in general, and ameliorate the drudgery of women in particular. Mrs. Suman Krishan Kant, President, Mahila Dakshita Samiti also graced the occasion and said, "women are most exploited workers and they should come forward to solve their problems". Mrs. Krishan Kant released a book titled "Jottings in Human Resources Management" which was based on feedback and valued suggestions of the participants. Eight women scientists and managers from all over the country participated in the programme.

6. Production of Educational Videotapes

The course exposed the participants to the concepts of educational video programmes, and the creative aspects of them. It also developed the skills for programme production, and introduced the participants to the new and emerging areas of video application.

7 Management of Improving and Assessing Agricultural Research Productivity

The course intended to enable the trainees to understand and practice various skills needed for improving the research performance and productivity, and to acquaint them with the techniques to identify relevant farm problems for research and the methodology for generating appropriate technologies.

8 Training of Trainers

The programme helped the teachers and trainees to improve their knowledge and develop certain specialized skills in order to organize and conduct more effective training programmes. The educational technology utilized in the programme consisted of a blend of lectures, exercises, group discussions and practice sessions to cover various aspects such as basics of training, role of trainers, and training design, delivery and evaluation.

9 Faculty Development Programme on Educational Technology

This programme was intended to introduce various instructional systems and strategies and instructional methods for optimizing teaching-learning effectiveness in the University. The new concepts and approaches for effective instruction, thus introduced, were used to prepare instructional materials in the teaching programmes of the University.



RESEARCH

The Academy is endowed with mandate to conduct research in the area of agricultural research management. It carries out research in various areas such as agricultural policy, research project management, transfer of technology, human resources development, agricultural education, and information technology. A brief description of the research projects is presented below :

1. A profile of Scientific and Technical Human Resources in NARS
(Dr S. Shanmugam and Dr Jagdeesh C. Kalla)

The project was undertaken to study the socio-economic attributes of cadre strength of scientific personnel in ICAR institutes, and to plan the optimum human resources profile in ICAR system for 2001 A.D.

The data have been collected. The compilation and statistical analysis of data were carried out. The report work is in progress.

2. Forecasting groundnut scenario
(Dr D. Rama Rao and Dr Kiresur)

The project was undertaken to forecast groundnut scenario in our country by 2000 A.D. using technological forecasting methods.

During the period under report, the review of global and Indian groundnut scenarios was taken into account along with the identification of institutions/agencies/individuals involved in each of the areas associated with groundnut. Brainstorming was conducted at the Academy to generate issues of importance for the project. The issues, thus generated, were utilized for the preparation of questionnaire. Three different questionnaires were designed for the scientists, administrators, and personnel in processing, marketing and post-harvest operations separately.

About hundred panel experts were identified from different sections of the society, out of which, seventy one persons agreed to be Delphi experts for collection of data. The analysis of the data received in the first round of Delphi is in progress.

3. **Forecasting the future scenario of flavours**
(Dr D. Rama Rao and Himanshu)

The main aim of this study was to forecast the future scenario of flavours in India with an emphasis on natural flavours.

The forecasts were made using Delphi, trend extrapolation and expert consultation methods. The study shows that the global flavour industry is expected to grow at the rate of 7-10 per cent in the coming decade, associated with the expected growth in food sector. It is also found that 70 per cent of the flavour market is restricted to USA, Europe and Japan regions.

The findings of the project indicates that the Indian share of global market comes through the export of essential oils, which is expected to show 7 to 9 per cent growth rate by 2000 A.D. This increase is mainly from the enhanced production of essential oils, some of which are produced only in India. In future, the beverage sector would be the highest consumer of flavours followed by confectionaries and dairy sector, both globally and nationally.

In India, the forecast predicts that, application of bio-technology would be highest in agriculture followed by food processing sectors. It is also interesting to note that in Indian flavour sector, application of biotechnology is almost nil at present. However, it will become a potential future tool largely due to demand from organised industrial houses.

4. **Agricultural Universities Information System (AGRIUNIS)**
(Dr D. Rama Rao and Mr U. Muralidhar)

The study aimed at collecting and analysing data of agricultural institutions, numbering 193 colleges in 27 State Agricultural Universities and 48 colleges in 26 other universities providing agricultural education in India. The data were collected through 5 different schedules designed to seek information on personal, academic and general, manpower, R & D and budget activities. A user-friendly software package was developed to maintain the database.

The project study made the following recommendations based on the experiences and interactions with officials of various agricultural uni-

versities. The study also put forward the action to be taken pertaining to a specific activity.

- a) For uniformity and periodic updation of data, the Planning and Monitoring Cells (PMCs) of the agricultural universities need to be strengthened and coordinated. Specially, PMCs need to be provided essential resources like manpower and associated infrastructure like computers. The cell needs to function directly under the vice-chancellor to facilitate active functioning and provide information from all wings of the university.

The PMC personnel need to be oriented to the importance of maintaining uniform databases across the universities. Since these personnel in SAUs are mobile, such training need to be provided periodically.

- b) To streamline the data collection based on schedules designed as per the requirement of the university, personal information schedule can be sent along with the annual assessment form. This will aid the VCs and Deans in rationalizing the faculty time for different activities.

- c) A co-ordinating body, like National Centre for Agricultural Information, is essential to monitor the progress and to organise the above mentioned programmes. Such a body can be established at the ICAR headquarters for strategic planning and NAARM can be associated for training and software development.

Besides these, there are several other important findings which can be utilized by all the agricultural universities. The project also recommended to take-up studies on issues of 'professional women in agriculture', 'Students drop-out in Agricultural Education', 'Manpower Planning for Agricultural Education and training needs of SAU's faculty.'

5. Development and analysis of database on Crops, Forestry, Animal and Fishery Resources in National Agricultural Research Systems

(Dr K. Vidyasagar Rao and Dr P. Kumar)

During the period under report, the data were compiled and updated on the parameters of livestock and fisheries enterprises like live-

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stock population, production data of whole milk and dairy products, meat from different species, number of animals sacrificed and total meat production, and data of wool, eggs, hides and skins, total catch, inland and marine resources, export and import of fisheries commodities. The project is in progress.

6. **HRD strategies for organizational effectiveness in National Agricultural Research System**
(Dr Jagannadham Challa, Mr U. Muralidhar and Dr Jagdeesh C. Kalla)

Keeping in view of the importance of human resources development for improving the effectiveness of research institutes through individual scientists, the project was proposed. The proposal was documented and submitted to ICAR for funding. A paper entitled, "HRD Strategies for Agricultural Research - An outlook" was prepared based on the preliminary work carried out so far to identify the areas of thrust for the research project. The work is in progress.

7. **Developing suitable Management Information System (MIS) for the National Agricultural Research System (NARS)**
(Dr T. Balaguru, Dr P. Manikandan and Dr Jagdeesh C. Kalla and Dr K. Vidyasagar Rao)

The study was undertaken mainly to develop a methodology for the development and use of a computer-based management information system (MIS) as a management tool for improving the managerial effectiveness in the research systems.

The basic MIS methodology including three case studies pertaining to ICAR institutes and agricultural universities were finalised. The report in the form of a book is in the final stage of publication. The developed methodology and the case studies have been extensively used in the MIS training programme organized at the Academy.

8. **Dynamics of agricultural management in public and private sector organizations - Practical implications and policy issues**
(Dr C. Prasad and Dr T. Balaguru)

The project was submitted to the ICAR for approval as adhoc research project under A.P. Cass Fund Scheme. The work was initiated in

anticipation of the final approval to collect information from agricultural organizations.

9. Development of Methodology for estimation of technological needs of farmers for estimation of farmers' problems
(Dr VE. Sabarathnam)

The research project was taken up to device clearcut step-by-step methodology to identify the technological needs of the farmers in order to formulate research and extension programmes for any discipline in agriculture.

The study recommended 14 steps for identification of the realistic problems and needs of the farmers. They are as follows:

- a) Identification of key informants.
 - b) Ranking the problems.
 - c) Estimation of economic importance.
 - d) Entering the ranks given by the key informants in a separate table, calculation of rank based quotient for each of the problems.
 - e) Working out the average loss for the various problems based on key informants' problems.
 - f) Selection of the farmers for identification of problems.
 - g) Collection of the technological needs and problems from the farmers who experienced the problems in realistic situation.
 - h) Quantification of the data, i.e., estimation of rank based quotient for each of the problems based on the farmers' responses.
 - i) Estimation of damage due to the technological needs of the problems in the actual situation.
 - j) Finding the correlation of the Rank Based Quotient and the average loss between the key informants and rest of the farmers using spearman Rank Correlation Technique.
 - k) Assessing the magnitude of the problem by multiplying the Rank Based Quotient value with average loss or extent of damage and multiplying the same with the area of the crop or no. of animals or birds as the case may be.
 - l) Separation of the research technological needs and extension problems.
 - m) Sending research technological needs to the research institutes for finding the solution.
 - n) Sending the extension problems to the development departments.
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EDUCATION

Outreach Programme on Educational Technology

Distance education has been accepted as a well recognised mode of education which is relevant to the existing demands of the society. It is a felt need to impart training to the academicians of State Agricultural Universities in the area of educational technology and management. To serve this purpose, the Academy has started the Outreach Programme in Educational Technology in 1991 to upgrade their knowledge, abilities, competencies and understanding in their respective fields of work or specialisation so that they can become more effective academicians and improve the general quality of agricultural education.

The programme is of one year duration with 46 modules, each module deals with specific class based issues under relevant curricular area of study. These modules are organised in five learning packages. The response sheets of each module, when completed and evaluated, facilitate the learners to get their learning reinforced. The programme has three contact sessions each of two weeks duration.

During this year, the third contact session of OPET-II was organised from June 6 to 13, 1994. Fourteen academicians of SAUs were awarded PG Diploma in Educational Technology. OPET-III was initiated during this year and the first contact session was conducted from March 27 to 31, 1995. Twenty academicians of various cadres from SAUs registered for the programme. The programme was coordinated by Dr A.Gopalram.

Summer Institute

Summer Institute on "Recent Advances in Agricultural Educational Technology" was organized at the instance of ICAR from May 7 to 26, 1994. The programme was designed to expose the faculty of various agricultural universities and ICAR deemed-to-be universities on different instructional systems, strategies and methods for enhancing teaching, learning effectiveness in agricultural universities.

Dr A. Ahmed, Dy. Director General (Education), ICAR, New Delhi inaugurated the programme. The training programme at the Summer Institute was a blend of structured experiences, group discussions, role plays and case studies. The programme included the topics like structured learning system, self-instructional methods, video programme production, video editing, computer assisted instruction etc.

The feed back from the participants indicated that all the participants liked the design and were satisfied with the organization of the summer institute. Most of them mentioned that the practical exercises helped them in developing new instructional skills and sharpening their existing skills in use of the instructional materials. They also appreciated the teaching methodology adopted on relevant topics.

In all, 25 Professors, Assoc. Professors of State Agricultural Universities and ICAR Institutes deemed to be universities representing various states participated in the summer institute.

Dr A. Gopalam was the Director of the Summer Institute.

Video Programmes on Agricultural and Allied topics

Some cassettes of educational video programmes, mostly discussion-oriented with suitable visuals (like video-clips, photographs, slides, tables etc.) on various agricultural topics have been produced by the agricultural scientists under training at NAARM. A list of the programmes produced is as follows.

1. Parthenium and its control
 2. Paneer-making
 3. Soil sampling techniques for soil testing
 4. Mush-rooms for meals
 5. Milk - the magic food
 6. Management of Nutritional disorder in groundnut
 7. Pests of Tomato
 8. Successful Quail-farming
 9. Integrated Nitrogen Management in rice
 10. Insect pest management in paddy
 11. Pashuonke Sankramak Rog (Hindi)
 12. Jaiv Urvarak - Ek Bemisal Vikalp (Hindi)
 13. Paddy-cum-fish culture
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14. Diseases of paddy
15. Calf-management
16. Mastitis in cows
17. Rabbit-rearing
18. Embryo-transfer in animal reproduction
19. Deemak - Ek Bhishan Samasya (Hindi)
20. Iron and Zinc Nutrition in Rice
21. Green manure in rice cultivation
22. Watershed management
23. Rice tissue culture
24. Induced Breeding
25. Aquarium maintenance
26. Soil Erosion
27. Pearl Culture
28. Nutrient management in sustainable rice production
29. Artificial brooding of chicks
30. Breeding insect resistant varieties of rice
31. Growing beautiful roses
32. Statistics in agriculture
33. Bio-control of rice pathogens
34. Dhan Phasal mein Iron Ka Abhav (Hindi)
35. Zero Energy Cool Chamber
36. Management of poultry hatchery
37. Hybrid sunflower production
38. Weed management in Groundnut
39. Epizootic Ulcerative Syndrome infishes
40. Gamee (Hindi)
41. Duck-rearing
42. Egg and its preservation
43. Agro-forestry
44. Pests and diseases of Sorghum
45. Anaj ka Bhandaran (Hindi)
46. Use mango-grafts for higher yields
47. Power tiller - boon for small farmers
48. Gulab (Hindi)
49. Your Indoor plants
50. Prerith Byanth - Induced Lactation (Hindi)
51. Artificial Insemination in cattle
52. Drip Irrigation
53. Striga - Devastating Witch-weed
54. Heat Detection in cattle

55. Azolla as bio-fertilizer
 56. Rearing of Chawky in Sericulture
 57. Aromatic plants
 58. Potash deficiency symptoms on crops
 59. Methods of rat-control
 60. Storage-bin for high moisture grains
 61. How to increase milk-production
 62. Ornamental orchards
 63. Computers and their use in agriculture
 64. Kitchen gardening
 65. BHV-Dairy Udyog ke liye Khatra (Hindi)
 66. Safe use of pesticides
 67. Role of rural women in agriculture
 68. Coccidiosis in Poultry
 69. Cacti and succulents
 70. Preventive maintenance of tractors
 71. Integrated Pest management in cotton
 72. Vitamins - the miracle medicine
 73. Epicotyl grafting in Mango
 74. Coleus - a medicinal plant
 75. Sesamum Improvement through wide hybridisation
 76. Control of water hyacinth
 77. Destruction of coconut and their management
 78. Hybrid capsicum cultivation
 79. Foot and mouth disease in animals
 80. Earn more through Jute
 81. Tissue culture for farmers' benefit
 82. Basics of layer-farming
 83. Bhuse Ki Urea Upachar (Hindi)
 84. Improved Techniques for groundnut cultivation
 85. Cultivation of water chestnut
 86. Management of buffalo calves
 87. Bonsai - an art
 88. Freshwater prawn-farming
 89. Hybrid rice
 90. Common fish-diseases of freshwater farms
 91. Management of nematode diseases
 92. Subabul - a promising tree
 93. Chickpea diseases and their control
 94. Infectious Bursal Diseases in chicken
 95. Cuscuta a parasitic weed
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96. Vermi-culture
 97. Neem evan uski upyogitahyein (Hindi)
 98. Artificical Hybridisation in Groundnut
 99. Makhana cultivation and processing
 100. Neem-coated urea for higher paddy yields
 101. Veneer grafting for better yields
 102. Bees and Beesflora
 103. Communication : Basic concepts
 104. Rabies
 105. Alley cropping for Agro-forestry
 106. Nutsedge and its control
 107. Rinder pest
 108. Flower Gardening
 109. Nutrition Deficiency - Sorghum
 110. Pests and diseases of Betelvine
 111. Plants poisonour to Animals
 112. Poshak - A weaning food
 113. Floral arrangements
 114. Suraj mukhi mein paragan samasya
 115. Cold water fisheries
 116. Khumb ke kheti
 117. Cotton pests and their management
 118. Broiler Rearing
 119. Micro propagation of forest trees
 120. Oyster mushroom cultivation
 121. Marine acquarium fishes
 122. Ganne ke krishi yantra
 123. Debeaking in poultry
 124. Clean meat production
 125. Culturable varieties of Fish
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WORKSHOPS/SEMINARS

The workshops/seminars/conferences were organized by the Academy to facilitate exchange of ideas, to disseminate information, and also to pool information resources among the scientists and research managers at National Agricultural Research System (NARS). The objectives of such workshops, conferences conducted were centred around all aspects related to the agricultural research management. The list of workshops conducted during the period under report is given in the table.

1. Workshop on "Management Information System for Agricultural Research"

The three-day national workshop was organized for the senior executives from the ICAR institutes and State Agricultural Universities to expose to the MIS Methodology developed by NAARM. Several Management issues associated with the implementation of MIS were discussed besides sensitizing the participants for the need to develop computer-based Management Information System for the effective implementation of MIS in the research system. The recommendations emerged from this workshop were circulated to all ICAR institutes and SAUs for follow-up action. Thirteen Directors from ICAR institutes participated in the workshop.

2. Mobile Seminar on "Management of Training - Local Institutional Resources and Experiences"

This unique mobile seminar was conceived and implemented by NAARM to share and experience facilities in terms of physical, academic and methodological possessed by the agricultural institutions located in and around Hyderabad. The one-week long seminar deliberations paved the way to create awareness of various physical facilities, to understand the innovations and pedagogy of training, to know the expertise available, and also to chalk-out a strategy for linkages between these institutions. Eleven institutes in the Hyderabad participated in the Seminar.

As an outcome, a directory of resource persons and physical resources, and a catalogue of periodicals in 20 libraries has been prepared.

LIST OF WORKSHOPS / SEMINARS

Sl. No.	Name of the Programme	Dates	Coordinators	No. of participants
1.	Workshop on Management Information System for Agricultural Research	Apr.19-21 1994	Dr T. Balaguru Dr Jagdeesh C.Kalla	13
2.	Mobile Seminar on Mgmt of Training-Local Institutional Resources and Experiences	June 27- July 2 1994	Dr C. Prasad Dr M.M. Anwer	11
3.	Workshop on Participatory Rural Appraisal and Participatory Technolgy Development	July 16 1995	Dr A.U. Gadewar Dr R.K. Samanta	30
4.	Workshop on Non-Pesticidal Approach to Pest Management - A New Direction	Sept.20-22 1994	Dr C. Prasad Dr R.K. Samanta	200
5.	Regional Conference of Peasant Women of Southern States in Agriculture, Environment and Rural Development.	Oct,27-29 1994	Dr C. Prasad Dr VE. Sabarathnam	100
6.	International Training Workshop on Management of Agricultural Research	Nov. 21 - Dec.2 1994	Dr C. Prasad Dr J.C. Kalla Dr T. Balaguru Dr M.M. Anwer	18
7.	Workshop on Manpower Planning and Development in Extension Education for Meeting Future Challenges of Agriculture	Feb.13-16 1995	Dr C. Prasad Dr R.K. Samanta Mr A.N. Shukla	114

3. Workshop on "Participatory Rural Appraisal and Participatory Technology Development"

The Academy after intensive discussions by faculty members have decided to incorporate Participatory Rural Appraisal (PRA) in field Experience Training (FET) of the ARS probationers. Several participatory methods of data collection for research problems identification are employed using PRA methodology. Keeping in view of the significance of PRA in FET, day-long in-house workshop was organized to develop insights into the methods and methodology, to develop multi-disciplinary perspective, and to develop necessary skills to employ PRA methods and methodology in management areas. Thirty faculty members and officers of the Academy attended the workshop. Mr E. Venkataramaniah, Exe. Dir., Youth for Action, Hyderabad was the resource person.

4. Workshop on 'Non-Pesticidal Approach to Pest Management - A New Direction'

The three-day workshop was conducted in collaboration with Department of Agriculture, Govt. of A.P., Centre for World Solidarity and OXFAM (India) Trust, with an objective of assessing the work on non-pesticidal approach and experiment with other major pests. The workshop also discussed the steps necessary for successful group action by farmers in extended areas of Andhra Pradesh, and the role of indigenous knowledge and farmers participation in scientific activities with a view to develop guidelines for future work in pest management.

Hon'ble Governor of A.P., Mr. Krishan Kant, while inaugurating the workshop, called for developing effective non-pesticidal techniques for pest management in view of ecological balance.

An exhibition on chemical, pesticide-free, methods of pest management was organised which was visited by hundreds of farmers, both men and women, from several districts of Andhra Pradesh. A KISAN GOSHTI was also organized based on the objectives of the workshop. This Goshti formed a common platform for a fruitful collaboration between government organizations and non-governmental agencies. The farmers in the KISAN GOSHTI shared their experiences with scientists of local ICAR Institutes.

About 200 leading agricultural scientists, officials and NGO activists participated in the workshop.

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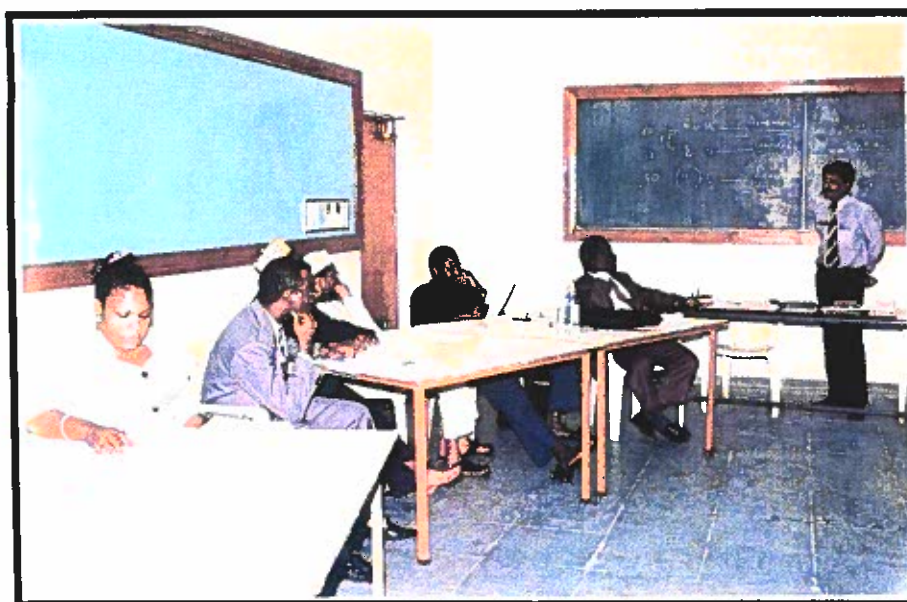
ARS probationers in action at Participatory Rural Appraisal during FET.



Peasant women voicing their experiences and problems during the regional conference of "Peasant Women of Southern States in Agriculture, Environment and Rural Development".



The faculty members of Haryana Agricultural University engrossed in group discussion during Faculty Development Programme in Educational Technology organized by the Academy at Hissar.



The Nigerian delegates, who participated in an international training workshop on "Management of Agricultural Research", are interacting with Dr Jagannadham Challa, Sr. Scientist, NAARM.

5. Regional Conference of "Peasant women of Southern States in Agriculture, Environment and Rural Development"

The three-day Regional Conference of Peasant Women was organized with a main objective to promote greater interactions between peasant women, and scientists, technologists and representatives of government agencies involved in shaping the policies for agriculture, environment and general rural development and to strengthen the existing programmes for poverty alleviation and empowerment of rural women such as IRDP, DWCRA, Jawahar Rozgar Yojana, STEP, Mahila Samakhya etc. through the process of collaboration that these conferences may help to initiate. The conference was organized in collaboration with the Centre for Women's Development Studies (CWDS) and the International Federation for Women in Agriculture (IFWA).

The conference which was of informal nature, facilitated the farm women to interact with the agricultural scientists and with the representatives of women organizations candidly.

The deliberations reflected the problems and status of the farm women in different states of South India. The outcome of the conference form the ground work for the extension workers, government departments, non-government organizations and other voluntary agencies working for the upliftment of the rural women. These problems will be borne in mind by all concerned in future whenever development programmes are designed.

The farm women belonging to different states of South India expressed some of the following problems:

- a) Wage disparity between men and women labourers.
- b) Lack of timely advice on modern methods of agriculture as they do not know much about timely operation, alternate cropping system and other aspects related to farm production cycles.
- c) Inspite of the various schemes and development programmes and the concern of the international agencies and FAO for the welfare of the farm women, these schemes do not reach the needy farm women.

About 100 landless agricultural labourers and farm workers from Andhra Pradesh, Karnataka, Kerala, Tamil Nadu and Pondicherry took part in the conference.

6. International Training Workshop on "Management of Agricultural Research"

An international training workshop on the management of agricultural research, for the Directors of Agricultural Research Institutes of Nigeria, was organized for two-weeks in collaboration with the Agricultural and Rural Management Training Institute (ARMTI), Nigeria.

The workshop was administered in two phases. The first phase was held at ARMTI, whereas, the second phase was held at NAARM. The programme schedule was drawn up with a focus on training in the management issues of agricultural research.

All the interactions were through different technical sessions dealing with the strengths of National Agricultural Research Systems (NARS) of India and Nigeria; agricultural research planning; research and administration interface; research project management and management information system; human resource management; Indian experience with the National Agricultural Research Project (NARP); extension education; adaptive research; international perspective to management; private sector R & D, and computer applications in agriculture. A special session on post-harvest technology and food processing was also arranged.

Several top level experts in the country were invited to interact with the participants on all the above mentioned subjects. Besides discussions, field visits were also arranged to various agricultural R & D institutes in different parts of the country to acquaint with the latest developments of agriculture in our country.

At the end of the workshop, it was decided to further strengthen the collaboration between NAARM and ARMTI in the area of agricultural research management by taking up joint training programmes and case studies. It was also proposed that both the institutions might work out a memorandum of understanding with the concurrence of the appropriate authorities in Nigeria and India for close cooperation and exchange. The following seventeen Directors from Nigeria took part in the workshop.

Dr E.K. Okaisabor
Dr S.M. Misari
Dr J.S.O. Ayeni

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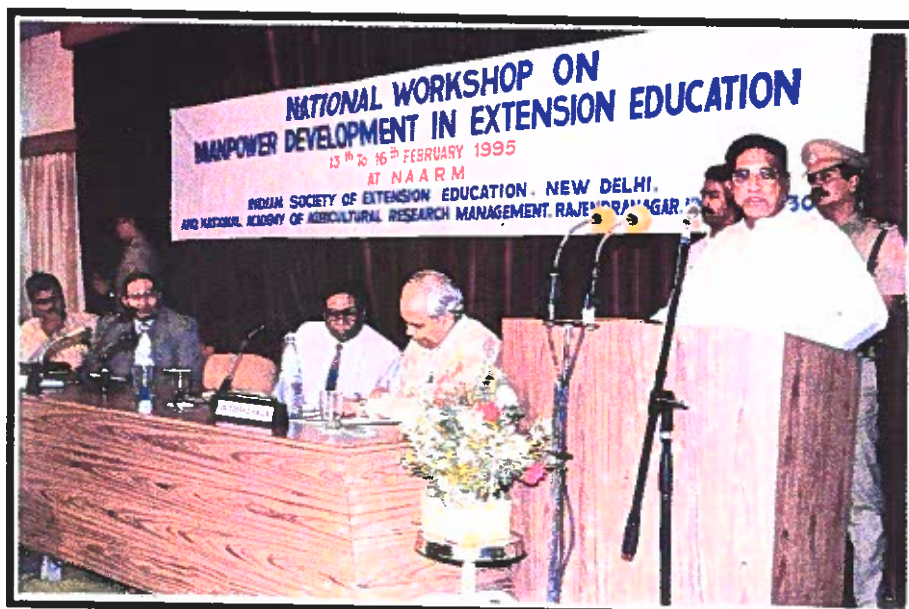
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His Excellency, Dr M. Channa Reddy, Governor of Tamilnadu, presenting the ISEE (R&D) Linker Award to Dr R.S. Paroda, Director General, ICAR.



Dr N.G.P. Rao, former Chairman, ASRB distributing certificate to Mrs M.I. Udo-Aka, Dy. Director, NARP, Nigeria in the valedictory function of international training workshop on "Management of Agricultural Research".



Dr M. Channa Reddy, Hon'ble Governor of Tamilnadu delivering the inaugural address during the national workshop on "Manpower Development in Extension Education". To his right (R to L) : Dr Jagdeesh C.Kalla, Jt. Director; Dr C. Prasad, Director, NAARM; Dr R.S.Paroda, Director General, ICAR; Mr M. Sashidhar Reddy MLA, AP.



Mrs Suman Krishan Kant, President, Mahila Dakshita Samiti, AP releasing a book titled, "Jottings in Human Resources Management" during national training workshop on "Human Resources Management for Women Scientists", flanked by Mrs Margaret Alva, Union Minister of State for Personnel and Dr C. Prasad, Director, NAARM.

Prof. L.B. Olugbemi
 Prof. Oyedipe
 Prof. B.O. Adelana
 Alh. M.B. Zaria
 Dr D.O. Ataga
 Mr J.G. Tobor
 Mr S.K. Onweluzo
 Mrs Udo-Aka
 Mr Agboola
 Dr Halid
 Prof. Odunfa
 Dr Olu Makinde
 Dr A. Adewumi
 Dr Adeyemi

7. Workshop on "Manpower Planning and Development in Extension Education for Meeting Future Challenges in Agriculture"

A four-day national workshop was organized in collaboration with Indian Society of Extension Education (ISEE) with the main objective to discuss, review and assess the future needs of manpower, and the quality of training that is required to meet the future challenges of agriculture.

Dr M. Chenna Reddy, Hon'ble Governor of Tamil Nadu facilitated the conduct of inaugural session. During his speech, he expressed concern over the escalating population which is main factor for diminishing the development rate.

The other objectives on which the workshop emphasized were: 1) to assess the developments in agriculture and agricultural extension 2) to understand the critical gaps in manpower development in agricultural extension 3) to work on the identified gaps and develop syllabi for the academic programmes for the degree courses as well as varied training programmes for extension scientists and the workers 4) to identify the critical gaps in the subject-matter specialist 5) to study the needs of the institutions and organizations offering degree or training courses in agricultural extension in terms of manpower and material supports, and 6) to study the latest trends and methodology being followed in our universities, colleges and training institutions.



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On the occasion, Dr Channa Reddy presented the ISEE Research and Development Linker Award to Dr R.S. Paroda, Director-General of ICAR, Dr M.V. Rao, Vice-Chancellor, APAU and Dr B.R. Mulik, Chairman, Threeway Agrotech, Pune and the ISEE Excellence in Extension Award to Dr Tito E. Contado, Chief of Agricultural Education and Extension Services, FAO, Rome. Dr P.N. Jha, Former Vice-Chancellor, Rajendra Agricultural University, Bihar and Dr A.W. Van Den Ban, Consultant on Extension Education, Netherlands. The International Federation for Women in Agriculture Award was presented to Ms Nirmala Deshpande, President, Bharatiya Harijan Seva Sangh, Delhi.

One hundred and fourteen scientists and professionals in the field of agricultural extension and transfer of technology from ICAR institutes, SAUs and other government and non-government organizations participated in the workshop. Extension professionals from Netherlands, Bangladesh, Australia, FAO, Rome and Italy also took part in this Workshop.

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CONSULTANCY

The Academy provides consultancy services in the area of Agricultural Research Management and Educational Technology. The Academy guides the SAUs and ICAR institutes in conducting training programmes in these areas. The faculty members of the Academy, specialised in various sub-disciplines, deliver special lectures in institutions which organise training programmes for their own staff. These institutions find the participation of NAARM faculty immensely useful.

The following consultancy services were provided by the Academy during the period under report:

1. FDP for HAU, Hissar and YSPUH&F (Haryana Agricultural University, Hissar and Y.S. Parmar university of Horticulture and Floriculture)

At the request of the Vice-Chancellors of HAU, Hissar and Y.S. Parmar university of Horticulture and Floriculture, FDPs on "Educational Technology" were conducted in Jan. 12-24, 1995 and July 2-8, 1994 respectively.

The programme was designed to introduce various instructional systems and strategies and instructional methods for optimizing teaching-learning effectiveness in the university. Dr A.Gopalam coordinated these programmes.

2. Consultancy to M/s.Nippon Koei Co.Ltd., Japan

Consultancy was extended through M/s.Nippon Koei Ltd., Japan to Andhra Pradesh Irrigation Projects for developing systematic crop plans for K.C. Kannal command area based on the local specific situations.

3. Consultancy to AP Irrigation Project

Consultancy was provided to III AP Irrigation Project through Water and Power Consultancy Services (Govt. of India undertaking) for developing cropping systems for command area of Srisailem right canal project and Srirani Sagar Project. Dr. K.Venkateswarlu developed feasibility of crops, crop calendar and cropping system based on the location specific needs of the farming community.

4. Consultancy to other training Institutes

The Academy provided expertise in the form of guest lectures to other local training and educational institutions such as the Extension Education Institute, the Home Science College of the Andhra Pradesh Agricultural University, S.V.P National Police Academy, Institute of Cooperative Management, Regional Station for Forage Production and Demonstration, International Crops Research Institute of Rural Development, WALAMTARI, NISIET, and NIRD. Most of the faculty members of the Academy were invited to deliver guest lectures in the above mentioned institutions. Subjects covered in these lectures were from the different areas of management and educational technology.

MISCELLANEA

Official Language Implementation activities:

Hindi Week Celebrations

Hind Week was celebrated from Sept.7 to 14, 1994 under the chairmanship of Dr.C. Prasad, Director, NAARM. The Chief Guest of the concluding function on Sept.14, 1994. Dr Shivendra Kishore Verma, Vice-Chancellor, CIEFL, Hyderabad emphasized the need for absorbing as many words as possible from other languages into Hindi to make it more live and powerful language. Dr C. Prasad, Director, in his presidential address said that one should develop affection towards one's own national language, and Hindi must be made easy to enable more people to learn without much difficulty. Dr J.C. Kalla, Joint Director and Chairman, Official Language Implementation Committee of the Academy said, "the culture of a nation is nurtured by the language of the nation. Hindi is the language that connects the entire nation as majority of our people speak Hindi". Dr A. Gopalam, Officer-in-charge, Hindi and Vice-Chairman of Official Language Implementation Committee of the Academy gave detailed account of various activities and achievements of Hindi section. On this occasion, various competitions like essay writing, elocution, typing, translation of words, noting & drafting, recitation of poetry, singing, etc., were organised and prizes were distributed to the winners.

Hindi audio cassette "SNEH" (Simple Natural Elementary Hindi) was released as a teaching aid to non-Hindi speaking trainee scientists of the Academy.

Workshops

Two workshops in Hindi were conducted from November 4th to 6th, 1994. First workshop was for strengthening the knowledge and skills of the employees of the Academy in the use of language and the other on the simplification of scientific language for extension of Agricultural Research. Another workshop was organised for the employees of the Academy during the period under report.

Rajbhasha Shield to NAARM

Rajbhasha Rolling Shield was awarded for the second successive year to the Academy in recognition of the appreciable work done in implementing the official language policy of the union during the year 1993 by the Town Official Language Implementation Committee (TOLIC) of Hyderabad in its 22nd meeting held at Railway Officers Club, Secunderabad on June 28, 1994. Dr. Jagdeesh C. Kalla received the shield on behalf of the Academy.

Mr Phanikar, Dy. Director, Official Language Implementation Office (Dakshin), Bangalore inspected the implementation of the official language policy and rules of Govt. of India for various official purposes by the Academy. He expressed great satisfaction regarding the use of Hindi for different purposes like correspondence, issuing circulars, publishing NAARM Newsletters and Annual Report, conducting typewriting training to the employees, imparting training to the trainee scientists, etc.

Sadhbhavana Diwas

The Academy observed Sadhbhavana Diwas on August 20, 1994. The staff members assembled and took a pledge to preserve the harmony and unity of the nation.

Award

Dr C. Prasad, NAARM Director has been associated with the Krishi Vigyan Kendras (KVKs) establishment in the country. In recognition of his achievements and contributions in the field of agricultural extension, the Chandra Sekhar Azad University of Agriculture and Technology conferred award of D.Sc degree (Honoris Cause) on Dec. 2, 1994 at its fifth convocation.

Rewards for case studies

The Venkateswara Trust instituted (Regd. No. 327/95 AP) at NAARM by Dr K. Venkateswarlu, Professor, NAARM governed by eminent personalities of ICAR and SAUs as executives, awards Rs. 20,016 (Rupees Twenty thousand sixteen only) to the scientist / manager who writes the best case research paper on Management of Agricultural Research Stations. The cash award will be given once in two years.

FACILITIES AND SERVICES

The Academy has established the following laboratories to make the teaching - learning process more effective and to help the trainees learn effective use of mass media for dissemination of information.

Video Instruction and Production Laboratory

This Laboratory comprising two television studios and a Master Control Room, facilitates intensive training of scientists in the preparation of effective resource materials for telecasting. It helps in conducting role play exercises for personality development. Video Educational programmes are also produced. It is also useful for proper speech communication skills.

Audio-Visual and Photography Laboratory

This laboratory is equipped with all modern photographic and audio reprographic facilities, provides an excellent opportunity to gain experience in the preparation and use of audio-visual aids for effective presentation. These include Overhead Projectors, Epidiascope, Coloured Transparency - making units, Slide Projectors, Tape Recorders and Public Address Systems.

Speech and Mass Communication Laboratories

These Laboratories facilitate the scientists to record their programmes such as interviews, discussions and talks, specially for the rural listeners.

Organisational Behaviour Laboratories

Individual and group exercises in team-building, leadership and group dynamics conducted in these laboratories provide the trainees an opportunity to understand the various aspects of organisational behaviour including human thought, action and behaviour.

Self-aided Instructional Laboratory

This laboratory provides opportunity for the participants to undergo structured and programmed instruction under self-learning situations.

Automated electronic equipments and computerised instruction facilitate the process. A set of carrels with Telex Caramate imparting a tape slide programme, a VCP with monitor for a video tutorial, and a computer for a programmed instruction are used. Software developed for learning Plant Pigments/Chlorophylls is being evaluated for its effectiveness as a learning tool.

Library

The Academy has a specialized library with its collection focusing on agricultural research management, science policy, system studies, project management, futurology, forecasting, educational technology, teaching transfer system/policies, women studies, information management, research communication, computer/information technology, human resource development, administration etc. The library has a collection of more than 15,000 books and over 650 books are added during the year. The library subscribes to 163 foreign journals, 130 Indian journals and gets 60 journals and newsletters on gratis/exchange. Besides this, the library has a good collection of reports and other publications of various international organisations and also Foreign Tour Reports submitted by the ICAR scientists.

It has a CAT database using CDS/ISIS package ver 3.07 for its book collection and is producing computer printed card catalogue from this database. The CAT database will very soon be made available for patrons with a search interface like Heurisko. The library has also developed an FTR database for accessing the collection of Foreign Tour Reports of ICAR scientists. The library has a CD-ROM work station with a Hitachi CD-ROM drive and a DCM-TURBO 2000 PC-AT. The CD-ROM databases available with the unit are AGRIS (1989-1995); AGRICOLA (1980-1994) ; AGRI SEARCH (1990-1994) ; TROPAG AND RURAL (1975-1992).

The General Information Service Terminal of National Informatics Centre (GISTNIC), installed in the library, makes available basic information about India, and important national and international information, viz., agricultural/census/economic statistics, railway/air time table, recreational and tourist information. GISTNIC terminal at the Academy is extensively used for supplying statistical information to the scientist trainees, faculty and staff of the Academy.

A secondary publication titled "Agricultural Research Management Abstracts" (ARMA) is brought out by the Information and Documentation Unit.

Computer Laboratory

A computer laboratory has been established with specially designed furniture to accommodate the Mini Frame computer with 4MB RAM and 300 MB Hard Drive along with 20 personal computers. The system also constitutes of three dot-matrix printers, two near-letter quality printers of FX Series and one ink jet-printer. Four PC-AT-486 with 8MB RAM and 200/500 MB Hard drive alongwith SVEA colour monitors have been acquired to strengthen the areas like Artificial Intelligence/Expert systems, multimedia and geographic information system. CCD plate for computers and video projection has also been provided for effective use of computer practicals which helps for better demonstration and viewing.

These systems are mainly used to provide intensive training on computers to the participants from ICAR and State Agricultural Universities at different levels through various courses and workshops on Management Information System, Computer applications, Computer Aided Instruction and Library applications. Softwares like DOS, WINDOWS, UNIX, CLIPPER, WORD PERFECT, QPRO, LOTUS 1-2-3, REFLEX, dBASE, HARVARD GRAPHICS, PILOT STORY BOARD and CDS/ISIS, have been loaded in all the personal computers for effective use in training as well as for day to day activities. Further, they are put into extensive use for the preparation of data bases and development of library information system. The computers are also being used for the administrative work particularly in preparation of pay bills, accounting and budgeting, recruitment processing and for the secretarial work. Necessary training is also imparted to the staff on the use of application packages like Wordperfect, dBASE and Lotus with the aim of increasing the efficiency of their working conditions. There is a proposal to upgrade the Computer Laboratory to take care of the areas like Artificial Intelligence/Expert Systems, Multimedia and Geographic Information System. In order to achieve this, one PC-AT-486 and 5 PC-AT-386 along with necessary softwares are being procured.

Services

Administration, Finance and Accounts

This wing of the Academy is organized into five sections : Administration I deals with establishment matters; Administration II with cash and bills; Administration III with stores and purchases; Administration IV with estate management and training; and all matters pertaining to expenditure

and bank reconciliation statements, monthly and annual accounts, budget and revised estimates, and internal audit are dealt with by the audit and accounts section.

This wing provides administrative, financial, and accounts support to all the units of the Academy. Further, the Chief Administrative Officer and the Finance and Accounts Officer function as regular faculty members in delivering lectures on various rules and regulations of ICAR to ARS scientists and other trainees.

Farm Services

The Academy has developed the Agro-Technology Demonstration Blocks (ATDB) in an area of 30 hectares, which act as the show-window of the National Agricultural Research System. The concept of ATDB is a unique innovation where the technologies developed and released by different ICAR institutes and SAUs are demonstrated at one place for the benefit of national and international visitors. During the year under report, about 2000 visitors and 300 scientist trainees had an opportunity to view these demonstrations.

Production and Publication Services

The Production and Publication Cell undertakes publication of various materials developed by faculty for training purposes. The cell is equipped with Desk Top Publishing (DTP) System with several packages including MS Word, MacWrite, MacPaint, MacDraw, Page-Maker and Hindi Font. This cell is equipped with an offset printing press (Swift 300 Delux) processing camera, plate making, screen printing, and binding systems. The important publications brought out are NAARM News, Annual Report, NAARM Deepika, Recommendations of various Workshops, and ARMA (Agricultural Research Management Abstracts).

Hostel Services

The Academy's programmes are fully residential, and the training participants are accommodated in the Halls of Residence and in the Scientists Home. These residential facilities have excellent catering services, which provide Indian, Chinese, Mughlai and Continental cuisine. The Halls of Residence contain 154 fully furnished single occupancy rooms with attached bath. The Scientists Home has 32 suites. Each suite is furnished

with two beds, sitting and study areas and an attached bath. Half of these suites are air-conditioned. There is a public call office in the Halls of Residence with STD/ISD facility.

Banking Service

The Central Bank of India, Golconda Fort Branch, Hyderabad has been operating an Extension Counter (regular) in the faculty-cum-laboratory building to cater to the banking requirements of faculty members, staff and trainees. the counter also offers safe deposit locker facility to the faculty and the staff members.

Health Services

The Academy has a Health centre with a doctor and a compounder. The centre provides out-patients care to all probationers, faculty, staff and their families during scheduled hours. Emergency care is provided round-the-clock.

Public Relations Cell

The Public Relations Cell continued to engage in activities like effective linkage with the media to focus attention on the activities and achievements of the Academy. The cell, besides building the image of the Academy, also provides faculty support in the areas of publicity, visitors coordination, graphics art aid and arranging all Academy guest lectures. The cell also arranges cultural programmes, field trips and sightseeing programmes.

Other facilities

Residential Facilities

The Academy has a total of 95 residential quarters of different categories for the purpose of accommodating scientists, technicians and staff of NAARM and other ICAR institutions in Hyderabad.

Recreation and Sports Facilities

The recreation facilities provided include several televisions connected to a dish antenna, common rooms with indoor games, and courts for tennis, volley ball and badminton.

TECHNICAL INFORMATION

Financial Resources

Head of Account	Actuals during 1994-95			Budget grant 1995-96		
	Non-Plan	Plan	Total	Non-Plan	Plan	Total
Pay & Allowances	88.32	24.04	112.36	112.21	100.00	212.21
Travelling Expenses	1.58	8.90	10.48	1.60	12.00	13.60
Recurring Contingencies	51.46	11.76	63.22	29.19	60.00	89.19
Non-recurring Contingencies						
1. Equipment	--	38.66	38.66	--		
2. Works	--	36.69	36.69	--	103.00	103.00
3. Others	--	--	--	--	--	--
TOTAL	141.36	120.05	261.41	143.00	275.00	418.00

Human Resources

Sl.No	Name of the Post	Sanctioned Strength	Posts filled as on 31-3-95
1.	Scientific	43	20
2.	Technical	42	40
3.	Administrative	49	49
4.	Auxilliary	26	26
5.	Supporting	37	37
Total		197	172

96
otal

2.21

3.60

19.19

3.00

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3.00

PUBLICATIONS

Books

1. Samanta, R.K. (ed.) 1994. "India's North-East - The Process of Change and Development", B.R. Publishing Corporation, Delhi.
2. Pratap Reddy, K. and Anwer, M.M. 1994. "Interpersonal Trust and its causes : Evidence from some research organizations" *Indian Journal of Industrial Relation* 30:2:156-174.
3. Samanta, R.K., 1995. "Women in Agriculture" - Perspective, Issues and Experiences (Ed.) in press.
4. Samanta, R.K., Prasad, M.V. and Vanisri T. 1995 - Research studies in Agricultural Extension in India 1957 - 1993 (ed.) in press.

Research papers / Review papers / Popular articles

1. Balaguru, T. 1994. "Faculty Development strategies for Agricultural University Teachers". In : *Training and Development*, 24(1) : 33-46.
 2. Samanta, R.K. 1994. "Women in Agriculture". *Interpaks Digest*, Vol.2. No.2 & 3.
 3. Gadewar, A.U. 1994. "Preference of VEW's" *Interpaks Digest* Vol 2. No. 2 & 3:5-6, Summer/Fall, Urbana Champaign, Illinois, USA.
 4. Shirly Raichal Samuel; Sobhakumari, V.P.; and Gadewar, A.U. 1994. "Agro-Forestry Potential in Northern Kerala", *Kurukshetra* Vol.XLIII, No.3, December, 1994 : 24-28.
 5. Gadewar, A.U. 1994. "Importance of Priority Setting in Agricultural Extension", the *Hitwada*, Maharashtra.
 6. Rasheed Sulaiman, V; and Gadewar, A.U. 1994. "Privatisation of Extension Services - Implication in the Indian Context". *Journal of Rural Reconstruction*, XXVII (2) : 41-47.
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7. Behera, U.K., and Gadewar, A.U. and Satpathy, S. 1994. "Scientists Perception about Different Facets of Agricultural Research Projects", Maharashtra. Journal of Extension Education, XIII, 91-94.
8. Pratap Reddy, K. and Anwer M.M. 1994. "Interpersonal trust and its causes : Evidence from research organizations". In : Industrial Relations, Vol. 30, No.2.
9. Sabarathnam, VE. 1995. "Manpower planning process in extension system for better results". In the souvenir of National Workshop on 'Manpower Development in extension education for meeting future challenges of agriculture'
10. Samanta, R.K., 1994. "They reap less than they sow". The Hindu, Madras, April 18, 1994.
11. Samanta, R.K., 1994. "More women extension workers needed", Letter, International Agricultural Development, 14 (5), p.22.
12. Samanta, R.K., and Padaria, R.N., 1994. "Agriculture for Rural Development in India - An analysis", communicated for publication in Agricultural Situation in India.
13. Samanta, R.K., 1995. "Improving Women Farmer's access to extension services' Abstracts", Second agricultural science congress, National Academy of Agricultural Sciences, held at APAU, Hyderabad from January 19 to 21, 1995.
14. Samanta, R.K., and Prasad, M.V., 1995. "Training of extension personnel in India, Perspective of training participants, Abstracts", National Workshop on Manpower Development in Extension Education for Meeting Future challenges of agriculture held at NAARM, Hyderabad, from Feb. 13 to 16, 1995.
15. V.K. Mehan, P.M. Reddy, D. Mc Donald and K. Vidyasagar Rao 1994 "Components of Rust Resistance in Groundnut Genotypes Phytopathology", Vol.84 No.12: 1421-26
16. M. Vikram Reddy, P. Balasouri, V.P. Kiran Kumar, A.L. Cogle, K.P.C. Rao and K. Vidyasagar Rao 1994. "Sorghum Plant Growth in relation to

Earthworm Casting Activity in Semi-Arid Tropical Alfisols, Soil Biota" CSIRO Australia: 127-128

17. H.A. Van Rheenen, S. Bhatti and K. Vidyasagar Rao 1994. "The Sustainable Preservation of Biodiversity in Self Pollinating Plant Species". Sample Size and Collection Methodology. FAO/IBPGR Plant Genetic Resources Newsletter No.96,1-4.

18. R.K. Maiti, K. Vidyasagar Rao and G. Swaminathan 1994. "Correlation Studies between Plant Height and Days to Flower to Shortfly Resistance in Glossy and Non-Glossy Sorghum Lines". International sorghum and millet news letter no.35:109-11

19. V.K. Mehan, P.M. Reddy, D. McDonald and K. Vidyasagar Rao 1994. "Comparision of Whole Plant and Detached Leaf Methods for Studying Components of Rust Resistance in Groundnut.

Book Reviews

1. Balaguru, T. "An Analytical Approach to Problems of Indian Agriculture : A theoretical and systems approach" by B.H. Joshi. Productivity 35 (2) : 378-379, 1994.

2. Balaguru, T. "Animal Quarantine in Asia. The Pacific" by Asian Productivity Organization. Productivity 35 (4) : 750-751, 1995.

Technical Bulletins

1. Balaguru, T., Corbett, D.C.M., Nickel, J.L. and Ranawana, S.S.E. Panel Report of the External Review of ISNAR INFORM Programme held at ISNAR, Netherlands during November 21-23, 1994.

2. Prasad, C., and Balaguru, T. 'Women in Agriculture - Developmental Issues', Recommendations of the National Seminar on "Women in Agriculture - Developmental Issues", held at NAARM, Hyderabad during Dec.28-30, 1993.

3. Kumar, P., 1994. 'Agricultural Scenario of Nigeria'(Compilation)

Papers presented in Seminars, Conferences and Workshops:

1. Manikandan P. and Kalla, J.C. 1994. "Management Information System for Agricultural Research" in the Workshop on "Management Information System" held at NAARM from April 19 to 21, 1994.
 2. Challa, J. 1994. "Teacher's Role in Learner's Motivation", In the 'Faculty Development Programme in Educational Technology' held at Dr.Y.S. Parmar University of Horticulture and Forestry, Solan, Himachal Pradesh from July 2 to 8, 1994.
 3. Gadewar, A.U. and Rasheed Sulaiman, V. "Privatising farm extension - some issues", in the international workshop on 'Alternative and cost-effective extension approaches for sustainable agriculture : Methodological issues' held at TNAU from Dec.14 to 17, 1994.
 4. Gadewar, A.U., and Prasad, C. 1994. "Emerging Participatory Approaches - New Challenges to Extension" in the international workshop on 'Alternative and cost-effective extension approaches for sustainable agriculture : Methodological issues' held at TNAU from Dec. 14 to 17, 1994.
 5. Anwer, M.M. 1994. "Management of Human Resources in Agricultural Research - Indian Experience" in the International Training Workshop on 'Management of Agricultural Research' at NAARM held on Nov.23, 1994.
 6. Challa, J. 1994. "HRD Strategies for Agricultural Research - An Outlook" in the International Seminar on 'Human resource needs for Change in R & D Institution', Jointly organised by WAITRO, UNESCO and NISTADS at International Centre, New Delhi from Feb. 15 to 17, 1995.
 7. Challa, J., Kalla, J.C. 1994. "World Trade Agreement and Trade Related Aspects of Intellectual Property Rights - Relevance to Indian Agriculture" in the national seminar on 'New Economic Policy and Perspective to Rural Development' from Feb. 23 to 25, 1995, held at NIRD, Hyderabad.
 8. Gopalam, A. and Kalla, J.C. 1994. "Instructional Function and Teacher Orientation in Agricultural Education" in National Development group seminar held at NCERT, New Delhi from 28-29 March 1995.
 9. Gopalam A. 1994. "Lokoktiyon ke Avirbhaav Aur Krishi Sambhandhi Vigyanik Vishayon ke saralikaran par inka prabhav". "Bharateeya Tyohaar
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Aur Parvadin"; "Grameena Vikas Prabhandh Mein Paaramparik Madhyam Ka Prayog"; "Krishi Anusandhaan Prasaar Ke Liye Sthaneey Gyan Aur Uske Vigyaanik Aadhaar ko Saral Bhaasha Me Prastut Karne Ki Aavashaktaa"; "Krishi Vygyaanik Bhaasha ka saralikaran vanchit upaadaan"; "Hindi Madhyam se Diary Vigyaan Tathaa Krishi Vigyaan Kee Shikshaa"; "Sthaneey Aasthaayen Aur Enka Vygyanik Kaaran" was presented in the workshop on Krishi Anusandhan ke Lye Krishi Vygyanik Basha Ka Sarali Karan, held at NAARM from 13th to 17th December, 1995.

10. Sabarathnam, V.E. 1995. "Manpower Planning Process in Extension System for Better Results". In national workshop on 'Manpower Development in Extension Education for Meeting the Future Challenges in Agriculture' held at NAARM from Feb.13 to 16, 1995.

11. Balaguru, T; Manikandan, P. and Jagdeesh C. Kalla. "Management Information Systems for Agricultural Research" in the National Workshop on 'Management Information System', held at NAARM from April 19 to 21, 1994.

12. Manikandan, P. and Balaguru, T. "Agricultural Information Management System in India - An Overview". In the Fourth Regional Meeting of 'Statistical Database System', held at CGPRT Centre, Bogor, Indonesia from Oct. 12 to 14, 1994.

13. Balaguru, T. "Implementation of INFORM - Indian Experience". In the External Review Panel Meeting, held at ISNAR, the Hague, Netherlands during Nov. 21-23, 1994.

14. Challa, J. 1994. "Group Approaches for Optimizing Learning". In the summer Institute on 'Recent Advances in Agricultural Educational Technology' held at NAARM from May 7 to 26, 1994.

15. Challa, J., 1994. "Scientific Values and Professional Ethics in Agricultural Research". In the senior level programme on 'Management for Improving and Assessing Agricultural Research Productivity', held at NAARM, from Nov.21 to Dec.17, 1994.

16. Manikandan, P., Balaguru, T., 1994. "Management Information System for Agricultural Research". In the International Training Workshop on 'Agricultural Research Management' held at NAARM from Nov.21 to Dec.3, 1994.

17. Challa, J., "Personality Development of Teachers". In the 'Faculty Development Programme in Educational Technology', held at C.C.S. Haryana Agricultural University, Hissar from January 12 to 24, 1995.

18. Saha, S.N., Anwer, M.M., Manikandan, P., and Rama Rao, D., 1994. Status paper on "NAARM Resources and Training Expertise for Management Training" in Mobile Seminar on 'Management Training' held at NAARM, Hyderabad during June 27 to July 2, 1994.

19. Mohapatra, B.C., and Shanmugam, S., 1995. "ALP-ACP Ratio for Indicating Bio-chemical Stress in Tropical Fish *Lisa Parsia*. Abstracts". National Seminar on 'Current and Emerging Trends in Aquaculture and its impact on Rural Development' held at College of Fisheries (OUAT), Berhampur, Orissa during Feb 14-16, 1995.

VISITORS

A large number of distinguished personalities including national and international scientists visited the Academy during 1994-95. They include:

- ** Mr Bisham Marishana along with his three member team: Mr. S.B. Thapa, Mr.P.M. Shakaya, Mr.H.P. Srivastava - on April 4, 1994.
- ** Mr Upendra Varma, Member of Parliament - on April 20, 1994
- ** Mr Muneendra, Editor, "Dakshina Samachar" - on May 20, 1994
- ** Mr Kazuhito Matsushita, Head, Agriculture Division, Asian Productivity Organization, Japan - on July 9, 1994.
- ** Mr A.W. Vansenban, Netherlands - on July 22, 1994.
- ** Mr Adebayu Oncetan, ARMTI, Nigeria - on August 8, 1994.
- ** Mr Oladele Windapo, Asst. Chief Management Development Officer, ARMTI, Nigeria on August 11, 1994.
- ** Mr Wilson Ofage, Director, Crops Resources, Min. of Agriculture, Uganda and ten senior Ugandan and the senior Ugandan Extension Officials - on August 23, 1994.
- ** Fourteen officers of Agriculture, animal Husbandry, Horticulture, Sericulture and Fisheries Departments of Southern States - on September 14, 1994.
- ** Royal D. Coile, Cornell University and Jagdish Nazaveth Institute for Studies and Transformations, Ahmedabad - on Oct.3, 1994.
- ** Dr Derek Byerlee, GIS Specialist, World Bank - on Oct.6, 1994
- ** Twenty senior personnel from Afro-Asian countries - on Nov.1, 1994.

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- ** Twelve Ugandan senior officials - on Nov.14, 1994.
- ** Mr Yusuf Maigani, Commissioner, Ministry of Agriculture, Mr. Magaja D. Mohammed, Director, Agricultural Engineering and Mr. Ahmed Gamel, Director, Agricultural Services from Nigeria - on Nov.23'94.
- ** Seven-member delegation consisting of Directors of Extension from the Dept. of Agriculture, Sri Lanka were on study tour for 4 weeks from Dec.19, 1994 and had discussions with the faculty members of the Academy and also interacted with Krishi Vigyan Kendras to acquaint with the functioning of Agricultural Extension System in India.
- ** Dr N.B. Singh, Assoc. Professor (Forests), Dr.Y.S.P. University of Horticulture and Forestry, Solan, Himachal Pradesh - on Jan.9, 1995.
- ** Mr K.L. Jayatissa, Hordi, Dy. Director (Extn), Sri Lanka, Mr. Justise Nauwoga, Min. of Agriculture, Uganda and Dr.Md. Rezael Karim, Bangladesh Agril. University, Bangladesh - on Jan.10, 1995.
- ** Dr M.A. Kashem, General Secretary, Bangladesh Society of Extension Education - on Feb.10, 1995.
- ** B.Sc (Agri.) students from UAS, Dharwad, P.J.N. College of Agriculture, Konaikal University of Pondicherry; and College of Cooperative and Banking, Karnataka and Annamalai University during Feb.20-23, 1995.
- ** Seven-member high level delegation lead by Dr. Ismail Moharram from Yemen - on March 1, 1995.
- ** Mr. Bayani S. Agvirre, Sr. Dy. Resident Representative, UNDP - on March 7, 1995.
- ** Ten trainers of Management and Development Institutions from Asia- on March 10, 1995.
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NAARM ALUMNI ASSOCIATION

The NAARM Alumni Association serves the purpose of continuing the interaction of the Alumni with the Academy, and helps to keep alive the management perspective in their mind. The Association brings out a quarterly newsletter, and also encourages meetings in its local chapters. The office bearers of the Association are as follows:

PATRON : Dr. R.S. Paroda, DG,
ICAR

PRESIDENT : Dr. C. Prasad
Director, NAARM

GENERAL SECRETARY : Mr. R.C. Chelvan
Scientist (SG),
NAARM

Dr. M.M. Anwer
Sr. Scientist,
NAARM

VICE -PRESIDENTS (2)

Local : Dr. Vilas A. Tonapi
NRCS, Hyderabad

All India : Dr. A.K. Saxena
CIAE, Bhopal

JOINT SECRETARIES (4)

Eastern Region : Dr. K.K. Sharma
ILRI, Ranchi

Northern Region : Dr. Sanjeev Gupta
IGFRI, Jhansi

Western Region	Mr A. Chaturvedi NBSS & LUP, Nagpur
Central Region	Mr U.R. Badegaonkar CIAE, Bhopal
TREASURER	Mr B. Pandu Reddy Finance & Accounts Officer, NAARM

COMMITTEES**Academic Committee**

The ICAR has appointed the Academic Committee to study and review course curriculum for training, refresher/in-service training, seminars/conferences, international training programmes and areas of research. The composition of the committee is:

- | | | |
|----|---|------------------|
| 1. | Dr A.L. Chowdhary
Chairman, ASRB
New Delhi. | Chairman |
| 2. | Dr V.S. Vyas
Director, Institute of Development Studies
Jaipur. | Member |
| 3. | Dr D.P. Sinha
Former Director
Administrative Staff College of India
Hyderabad. | Member |
| 4. | Dr S.K. Sinha
National Professor
IARI, New Delhi. | Member |
| 5. | Dr A. Ahmed
DDG (Edn.)
ICAR, New Delhi. | Member |
| 6. | Dr K.S. Gill
Ex - Vice-Chancellor
Punjab Agri. University, Ludhiana. | Member |
| 7. | Shri G.S. Sahni
Secretary, ICAR, New Delhi. | Member |
| 8. | Dr Jagdeesh C. Kalla
Director
NAARM, Hyderabad. | Member-Secretary |

During the year under consideration, the committee met for five times to finalise the recommendations.

Quinquennial Review Team

The Quinquennial Review Team reviews the performance of the Academy with regard to achievement of its objectives in the last five years. It also examines and makes recommendation on the objectives, scope and relevance of the research programmes for the next five years in relation to the overall national plans, policies and long and short-term priorities. The team comprising the following members visited the Academy on Jan. 23, 1995.

- | | | |
|----|--|------------------|
| 1. | Dr R. Dwarakinath
Former VC, University of Agri. Sciences
FAO Advisor
Bangalore. | Chairman |
| 2. | Prof. S.C. Mandal
Former VC, RAU
Ranchi, Bihar | Member |
| 3. | Prof. N.S. Ramaswamy
Director, "CARTMAN"
Koramangala, Bangalore. | Member |
| 4. | Dr K.N. Nag
Former VC, Rajasthan Agri. University
Udaipur, Rajasthan. | Member |
| 5. | Dr R.K. Patel
Former-Director, NDRI
Karnal. | Member |
| 6. | Dr Dharni P. Sinha
Co-ordinator
Association of Management Development
Institutions of South Asia (AMDISA)
Hyderabad. | Member |
| 7. | Dr P.B. Mathur
ADG (Extn.)
ICAR, New Delhi. | Member-Secretary |
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Management Committee

The Academy is guided in planning, organizing, and conducting various types of training programmes and research studies by the Management Committee. The Management Committee approves the programmes to be undertaken. The Academy ensures their implementation, both in terms of physical targets and schedules, and provide broad guidelines for future programmes. The Management Committee also helps in formulation and consideration of proposals for Five Year Plan and Annual Plan and takes up periodical review of progress of development schemes.

The Management Committee, besides assisting Director of the Academy in the discharge his functions; focuses his attention on the research and other programmes of the Academy, and ensures their effective implementation. While evaluating these programmes periodically, it pinpoints bottlenecks, if any, and suggests remedial measures.

The 22nd, 23rd and 24th meetings of the Management Committee of the Academy were held on May 7, 1994; on October 4, 1994 and on February 11, 1995 respectively. The composition of the Management Committee of NAARM is as follows :

1.	Dr. C. Prasad Director, NAARM Rajendranagar, Hyderabad	Chairman
2.	Dr. Jagdeesh C. Kalla Joint Director, NAARM Hyderabad	Member
3.	Dr. S.D. Tripathi Director, CIFE Bombay	Member
4.	Dr. M.S. Chari Director, CTRI Rajahmundry Andhra Pradesh	Member

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| 5. | MrsK.Bharathi Devi
Social Worker
Kakinada, Andhra Pradesh | Member |
| 6. | Mr Shanta Kumar
Finance & Accounts Officer
CRIDA, Santhoshnagar
Hyderabad | Member |
| 7. | Mr. M.Suresh Kumar
Chief Administrative Officer
NAARM, Hyderabad | Member Secretary |

Academic Council

The Academic Council oversees all the academic functions of NAARM. The following was the composition of the Academic Council during 1994-95:

- | | | |
|----|---|----------|
| 1. | Dr. C. Prasad
Director, NAARM | Chairman |
| 2. | Dr Jagdeesh C. Kalla
Jl. Director, NAARM | Member |
| 3. | Dr P.Kumar
Principal Scientist, NAARM | Member |
| 4. | Dr R.K.Samanta
Sr.Scientst, NAARM | Member |
| 5. | Dr A. Gopalam
In-charge, EST Unit , NAARM | Member |
| 6. | Dr M.M. Anwer
In-charge, HRD Unit ,, NAARM | Member |
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| 7. | Ms Suguna Sri
In-charge, I&B Unit , NAARM | Member |
| 8. | Dr S.N. Saha
Principal Scientist, NAARM | Member-Secretary |

Research Council

The Research Council deliberates on the research activities in the areas of agricultural research and education management. The committee consists of the following members:

- | | | |
|----|--|----------|
| 1. | Dr C. Prasad
Director, NAARM | Chairman |
| 2. | Dr Jagdeesh C.Kalla
Jt. Director, NAARM | Member |
| 3. | Dr V.Raghavendra Rao
Principal Scientist, NAARM | Member |
| 4. | Dr P. Kumar
Principal Scientist, NAARM | Member |
| 5. | Dr S.N. Saha
Principal Scientist, NAARM | Member |
| 6. | Dr K. Venkateswarlu
Principal Scientist, NAARM | Member |
| 7. | Dr H.K. Samanta
Sr Scientist, NAARM | Member |
| 8. | Dr.A.Gopalam
Sr. Scientist, NAARM | Member |
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9.	Dr K. Purnachandra Rao Sr Scientist, NAARM	Member
10	Dr K.M. Reddy Sr Scientist, NAARM	Member
11.	Dr M.M.Anwer Sr Scientist, NAARM	Member
12.	Dr K. Vidyasagar Rao Sr Scientist, NAARM	Member
13.	Dr P. Manikandan Sr Scientist, NAARM	Member
14.	Dr Jagannadham Challa Sr Scientist, NAARM.	Member
15.	Dr S. Shanmugam Sr Scientist, NAARM	Member
16.	Dr VE. Sabarathnam Sr Scientist, NAARM	Member
17.	Dr D. Rama Rao Sr Scientist, NAARM	Member
18.	Dr A.U. Gadewar Sr Scientist, NAARM	Member
19.	Mr R.C. Chelvan Scientist (SG)	Member
20.	Mr U. Muralidhar Systems Analyst, NAARM	Member
21.	Dr T. Balaguru Sr Scientist, NAARM	Member Secretary

During 1994-95, four meetings were conducted by Chairman to finalise research programs of the Academy. In process, several projects pertaining the HRD, Project management, TOT and other areas relevant to Research Management were initiated.

Official Language Implementation Committee

The Official Language Implementation Committee of the Academy provides periodical suggestions for implementation of Official Language Policy, and rules and regulations of Government of India that are issued from time to time. The Committee met four times during the year on June 25, September 30, December 31, 1994, and March 31, 1995, and discussed several issues relating to the implementation of Hindi language. The following were the members :

1.	Dr C. Prasad, Director	Patron
2.	Dr Jagdeesh C.Kalla, Jt. Director	Chairman
3.	Dr A. Gopalram, Senior Scientist	Vice-Chairman
4.	Dr K.P.C.Rao, Senior Scientist	Member
5.	Dr D.Rama Rao, Senior Scientist	Member
6.	Mr M.Suresh Kumar, Chief Administrative Officer	Member
7.	Mr B. Pandu Reddy Finance & Accounts Officer	Member
8.	Mr K. R .Prabhakar Programme Officer (CCTV)	Member
9.	Mr U. Muralidhar Systems Analyst	Member
10.	Mr R.V.V.S.Prakasa Rao, Editor	Member

- | | | |
|-----|---|-----------------------|
| 11. | Mr Abhijit Bhattacharjee, PRO | Member |
| 12. | Ms Jhansi Laxmi, Sr.P.A | Member |
| 13. | Mr D.Venkateswarlu
Technical Officer (Hindi) | Member -
Secretary |

Training Cell

The training cell is responsible for all the training activities of the Academy. It maintains the record of all details of these programmes. Professor In-charge is also responsible for performance evaluation of the Foundation Course programmes. The training cell comprises of the following personnel:

1. Dr S.N. Saha, Professor In-Charge, Training Programmes
2. Mr P. Vijender Reddy, T-II-3
3. Ms Sarada Samanta Senior Stenographer

During the year under consideration, the training cell continued to plan the foundation and speciality courses. In all 12 speciality courses were conducted which trained 222 personnel from SAUs and ICAR Institutes. Similarly during 1994-95, 163 trainee-scientists were enrolled and completed the requirements of foundation course offered at the Academy.

PERSONNEL

1. Dr C. Prasad, Director (Till Feb.24, 1994)
Dr Jagdeesh C.Kalla, Joint Director (Director from Feb.25, 1995)

Agricultural Research Systems & Policies Unit

2. Dr P.Kumar, Principal Scientist & Unit In-Charge
3. Dr VE.Sabarathnam, Senior Scientist
4. Dr T.Balaguru, Senior Scientist
5. Dr K.P.C.Rao, Senior Scientist
6. Dr K.Vidyasagar Rao, Senior Scientist (on deputation to ICRISAT)
7. Ms A.Vijaya Lakshmi, Stenographer
8. Ms A.Mercy, Stenographer
9. Mr M.Shyam Rao, SSG-I

Educational Systems and Technology Unit

10. Dr A.Gopalam, Senior Scientist & Unit In-charge
11. Dr K.M.Reddy, Senior Scientist
12. Dr D.Rama Rao, Senior Scientist
13. Mr N.R.Nageswara Rao, Technical Officer (T-5)
14. Mr M.Venkatesh, Stenographer
15. Mr M.Yadaiah, SSG-II

Transfer of Technology Systems & Policies Unit

16. Dr R.K.Samanta, Senior Scientist & Unit In-Charge
17. Mr R.Chithiraichelvan, Scientist (Selection Grade)
(On study leave)
18. Dr A.U.Gadewar, Senior Scientist
19. Mr L.Venkateswarlu, Photographer (T-4)
20. Mr Bansidhar Nayak, Projectionist (T-II-3)
21. Mr D.Rajagopal Rao, PAE Operator
22. Mr S.Sunder Raj, Media Operator (T-1)
23. Ms T. Vanisri, Jr. Stenographer
24. Mr G.V.Bikshapathi, SSG-II

Project Management Unit

25. Dr V.Raghavendra Rao, Principal Scientist & Unit In-Charge
 26. Dr S.N.Saha, Principal Scientist
-

27. Dr K.Venkateswarlu, Principal Scientist
28. Dr P.Manikandan, Senior Scientist
29. Ms L. Jhansi Lakshmi, Senior Personal Assistant
30. Mr K.Satyanarayana, SSG-II

Human Resources Development Unit

31. Dr M.M.Anwer, Senior Scientist & Unit In-Charge
32. Dr Jagannadham Challa, Senior Scientist
33. Dr S.Shanmugam, Senior Scientist
34. Ms S.Sesha Sai, Junior Stenographer
35. Mr M.Narsinga Rao, SSG-I

Information and Documentation Unit:

36. Ms M. Sugunasri, Librarian, Head I&D Unit (From Feb.23, 1995)
37. Ms P.V. Nirmala, T-5 (Technical Officer)
38. Mr Y.V.Siva Prasad, Documentation Asst (T-4)
39. Ms S.Shanti, Junior Stenographer
40. Ms D.Dhanalakshmi, Lib.Asistant (T-2)
41. Mr K. Srinivasa Rao, Junior Clerk
42. Mr Venkatesham, Xerox Machine Operator
43. Mr M.Ashok, SSG-II

Director's Cell

44. Mr N. Raghunath, Stenographer
45. Ms K.K. Rukmani Ammal, Jr. Stenographer
46. Mr P.Swamy, SSG-II

Joint Director's Cell

47. Dr Jagdeesh C.Kalla, Jt. Director (Till Feb.24, 1995)
48. Dr Mohan Naidu, Jt. Director (From Feb. 23, 1995)
49. Mr T.V. Ramadas, Jr. Stenographer
50. Mr K. Hemlal, SSG-I

Training Cell

51. Ms Sarada Samanta, Senior Stenographer
 52. Mr P. Vijender Reddy, T-II-3
-

Hindi Section

- 53. Mr D.Venkateswarlu, Technical Officer (T-5)
- 54. Ms J.Renuka, Hindi Assistant
- 55. Mr S.Pradeep Singh, Hindi Translator
- 56. Ms P.Raja Rajeswari Jyothi, Junior Clerk

Computer Laboratory

- 57. Mr U.Muralidhar, Systems Analyst (T-7)
- 58. Mr M.J.Pradeep Kumar, Computer Operator (T-4)
- 59. Mr K.Venkateswara Kumar, Statistical Asst (T-II-3)
- 60. Mr P.Mohan Singh, Computer Assistant (T-II-3)

Video Instruction and Production Laboratory

- 61. Mr K.R.Prabhakar, Programme Officer (CCTV) (T-7)
- 62. Mr Ch.Janardhan Rao, Video Cameraman (T-4)
- 63. Mr N.Naresh Kumar, VCR-Tele Cine Operator (T-2)
- 64. Mr C.Bikshapathi, SSG-II

Production and Publication

- 65. Mr R.V.V.S.Prakasa Rao, Editor (T-6)
- 66. Mr N.Narasimha Reddy, Offset Machine Operator (T-5)
- 67. Ms G. Aneesa, Assistant Editor (T-II-3)
- 68. Mr K.Anil Kumar, Reprographic Assistant (T-II-3)
- 69. Mr P. Srinivas, Proof Reader
- 70. Mr Shekhar Reddy, Dark Room Assistant (T-2)
- 71. Mr S.Swamy, Assistant Gestetner Operator
- 72. Mr Prashant Gaikwad, Binder
- 73. Mr C.Chandramouli, SSG-I

Farm Section

- 74. Mr V.Murali, Farm Superintendent (T-6)(On Deputation)
 - 75. Mr K.Rangasamy, Field Assistant (T-4)
 - 76. Mr M.A. Basith, Lab Technician (T-4)
 - 77. Mr B.Veeraiah, Farm Assistant (T-I-3)
 - 78. Mr Ahire Laxman Mahar, Farm Assistant (T-2)
 - 79. Mr K.Shivaiah, Technician Electrical (T-1)
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80. Mr P.Bhaktavatsalam, Junior Clerk
81. Mr M.Padmaiah, Tractor Driver
82. Mr P.Iswari, Power Tiller Driver
83. Mr P.Balraj, SSG-II
84. Ms N.Sukunamma, SSG-I
85. Mr S.Jangaiah, SSG-II
86. Mr B. Subba Rao, SSG-II
87. Ms B.Bharatamma, SSG-I
88. Mr G.Pentaiah, SSG-I
89. Mr R.Sattaiah, SSG-I
90. Ms G. Anasuya, SSG-I

Hostel Services

91. Mr Zameer Ahmed, Manager (Hostel Services)
92. Mr Sham Bahadur, Catering-in-charge
93. Ms Savithri Srinivasan, Mess Supervisor
94. Mr C.J.Samuel, Junior Clerk
95. Ms D.Sangamma, Cook
96. Mr Khalid, SSG-II
97. Ms B.Santhamma, SSG-II
98. Ms S. Shakunthala, SSG-II
99. Mr A.Yadaiah, SSG-I
100. Mr M.Krishnaiah, SSG-I
101. Mr C.Anjaiah, SSG-I
102. Ms K.Kalavathi, SSG-II
103. Ms M.Yadamma, SSG-I
104. Ms I.Bharathamma, SSG-I
105. Ms T.Jangamma, SSG-I
106. Ms Nazia Parveen, SSG-I

Administration

107. Mr M.Suresh Kumar, Chief Administrative Officer
 108. Mr Mohammed Habeebuddin, Assistant Administrative Officer
 109. Mr K.Janaki Ramaiah, Superintendent
 110. Mr T.Ashok Kumar, Superintendent
 111. Mr Y.Sankara Rao, Assistant
 112. Mr P.P.Brahmaji, Assistant
 113. Mr K.S.Sharma, Assistant
-

114. Mr P. Neelakantam, Assistant
115. Mr P. G. Kohad, Assistant
116. Mr P. Anantha Reddy, Technician (Instrument) (T- II-3)
117. Mr K. Prabhudas, Senior Clerk
118. Mr M. Narasimha Rao, Assistant
119. Mr T. Srinivas, Senior Clerk
120. Mr M. Dinesh, Senior Clerk
121. Ms K. Swaraja Laxmi, Junior Library Assistant (T-2)
122. Mr A.M. Venkateswarlu, Compounder
123. Mr G. Raj Reddy, Senior Clerk
124. Ms N. Vijaya Lakshmi, Junior Stenographer
125. Ms B. Padma Saroja, Junior Clerk
126. Mr P. Srinivasu, Junior Clerk
127. Mr C. Phani Raj, Senior Clerk
128. Ms P.D. Jessie Ecclicia, PMBX Operator cum Receptionist
129. Mr M. Sridhar, Junior Clerk
130. Mr P. Venkatesh, Junior Clerk
131. Mr K.R. Ghanshyam, Junior Clerk
132. Mr M.K. Samson, Junior Clerk
133. Mr B.H. Dharma Raj, SSG-III
134. Mr J. Chandraiah, SSG-I

Typing Pool

135. Mr P. Anand Kumar, Stenographer
136. Ms V. Shailaja, Junior Stenographer

Public Relations Cell

137. Mr Abhijit Bhattacharjee, Public Relations Officer
138. Mr A. Murali Mohan Rao, Artist (T-4)
139. Mr P. Namdev, Graphic cum Visualiser (T-4)
140. Ms Y. Anuradha, Junior Stenographer

Civil Engineering Cell

141. Mr Sohail Ahmad Khan, Junior Engineer (Civil) (T-4)
 142. Mr K. Veeranarsaiah, Carpenter
 143. Mr N. Prabhakar Rao, Plumber
-

Transport Section

- 144. Mr K. Sattaiah, Driver
- 145. Mr S.K. Peera, Driver
- 146. Mr M.M. Mohan Rao, Vehicle Mechanic (T-3)
- 147. Mr G. Muthyalu, Driver
- 148. Mr U.V. Ratnam, Driver
- 149. Mr N.Ashok, Driver
- 150. Mr T.Laxman, Driver
- 151. Mr B.Narayana, SSG-III

Security Wing

- 152. Mr B.Ch.Satyanarayana, Security Officer cum
- 153. Mr G.Dasaratha, SSG-I
- 154. Ms M.Eswaramma, SSG-I
- 155. M.Phool Kumar, SSG-I
- 156. Mr I.Satyanarayana, SSG

Electrical Wing

- 157. Mr I.Ramesh, Electrician cum Plumber (T-I-3)
- 158. Mr K.Obulapathi, Technician Electrical (T-2)
- 159. Mr M.K.Sonkusare, Electrician (T-1)
- 160. Mr M.Mohan Rao, Technician Electrical (T-1)
- 161. Mr B.S.N.Murthy, Technician Electrical (T-2)
- 162. Mr M.K.Shamshuddin, Technician Electrical (T-2)
- 163. Mr A.Nageswara Rao, Lineman (T-1)
- 164. Mr B.Satyanarayana, Technician Electrical (T-1)
- 165. Mr D.R.Sahadeva Rao, Pump Driver

Audit and Accounts Section

- 166. Mr B. Pandu Reddy, Finance & Accounts Officer
 - 167. M. M. Girija Kumari, Junior Stenographer (on deputation)
 - 168. Mr C. Bagaiah, Assistant
 - 169. Mr S. Balakamesh, Junior Clerk
 - 170. Mr R. Chandra Babu, Junior Clerk
 - 171. Ms K. Radha Sujatha
 - 172. Ms Y. Gayathri, Junior Clerk
 - 173. Ms G. Manibai, SSG-III
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APPOINTMENTS

Sl.No.	Name	Designation
1.	Ms M. Sugunasri,	Librarian, T-6
2.	Dr Mohan Naidu,	Joint Director
3.	Mr P.Srinivas	Proof Reader
4.	Ms Dhana Lakshmi	Lib.Assistant

PROMOTIONS

Sl.No.	Name	Designation	Promoted as
1.	Ms A.Mercy	Jr Stenographer	Stenographer
2.	Mr P.Anand Kumar	Jr.Stenographer	Stenographer
3.	Mr M.Narasimha Rao	Sr.Clerk	Assistant
4.	Mr C.Phani Raj	Jr.Clerk	Sr.Clerk

DEPUTATION

1. Mr V.Murali (T-6) Farm Supdt. to T-7 Garden, Supdt at IIHR, Bangalore
2. Mr K.S.Sharma Assistant to Supdt. Directorate of Water Management Research, Rahur.

